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Supplement II
Examples on Doping and Fermi Levels

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This is a supplement presenting examples on the relationship between doping, carrier concentrations and the Fermi level at equilibrium.

Required Constants: For silicon at T=300 °K, $n_i = 1.45 \times 10^{10}$ 1/cm^3. The Boltzmann constant $k_B = k = 8.61 \times 10^{-5}$ eV/K. Silicon bandgap energy E_g=1.12 eV.

1. Consider a silicon crystal at room temperature (300 °K) doped with arsenic atoms so that $N_D = 6 \times 10^{16}$ 1/cm^3. Find the equilibrium electron concentration n_0, hole concentration p_0, and Fermi level E_F with respect to the intrinsic Fermi level E_i and conduction band edge E_C.

   This is an n-type material, as it is doped with donor atoms. Therefore:

   $$n_0 \approx N_D = 6 \times 10^{16} \text{ 1/cm}^3 \tag{1}$$

   Then we can use the Law of Mass Action to find the hole concentration:

   $$p_0 = \frac{n_i^2}{n_0} = \frac{2.1 \times 10^{20}}{6 \times 10^{16}} = 3.5 \times 10^3 \text{ 1/cm}^3 \tag{2}$$

   To find the Fermi level with respect to the intrinsic Fermi level, we use the expression that links electron concentration to E_i and n_i:

   $$n_0 = n_i \exp(\frac{E_F - E_i}{kT}) \;\Rightarrow\; (E_F - E_i) = kT \ln(\frac{n_0}{n_i}) \tag{3}$$

   At room temperature

   $$kT = 8.61 \times 10^{-5} \times 300 \approx 0.026 \text{ eV} \tag{4}$$

   We will be using this quantity often.

   Then the separation of the Fermi level and intrinsic Fermi level is, from Eqn. 3:

   $$\begin{aligned} (E_F - E_i) &= 0.026 \ln(\frac{6 \times 10^{16}}{1.45 \times 10^{10}}) = 0.026 \times 15.24 = 0.396 \text{ eV} \\ \Rightarrow E_F &= E_i + 0.396 \text{ eV} \end{aligned} \tag{5}$$

   Drawing the band energy diagram, we can then place the Fermi level in the correct place with respect to the intrinsic Fermi level (middle of bandgap) and also find its separation from the conduction band edge E_C. See Figure 1.

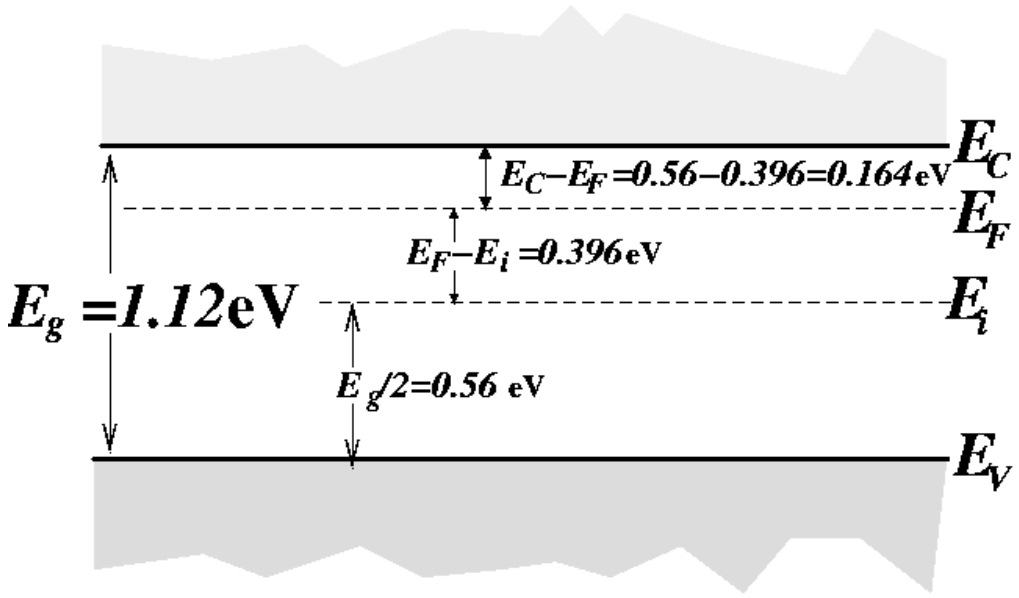


Figure 1: Solution to Problem 1. This is an n-type material doped at 6×10^{16} 1/cm^3.

2. Consider a silicon crystal at 300 °K, with the Fermi level 0.18 eV above the valence band. What type is the material? What are the electron and hole concentrations?

We start with drawing the band energy diagram for the crystal, as in Figure 2:

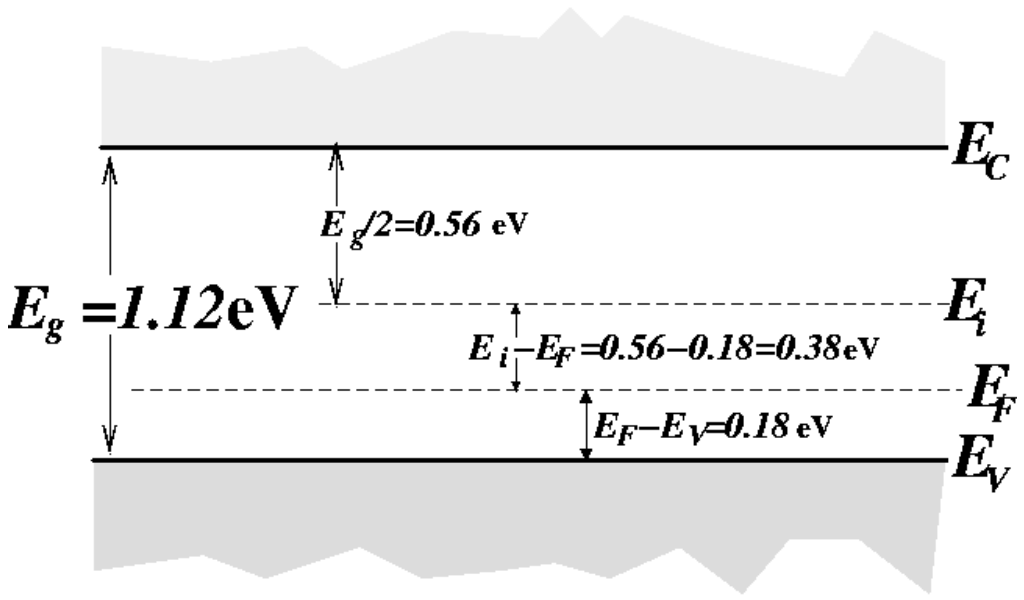


Figure 2: The energy band diagram for the crystal presented in Problem 2.

The separation between E_F and E_V is given. Then, as can be seen from the figure, we can calculate the separation between E_i and E_F:

$$\begin{aligned} E_i - E_V &= E_g/2 = 0.56 \text{ eV} \\ E_i - E_V = (E_i - E_F) + (E_F - E_V) \quad &\Rightarrow \quad (E_i - E_F) = E_g/2 - (E_F - E_V) \\ \Rightarrow (E_i - E_F) &= 0.56 - 0.18 = 0.38 \text{ eV} \end{aligned} \tag{6}$$

The Fermi level is below the intrinsic Fermi level (in other words, it is closer to the valence band than it is to the conduction band). Therefore this is a p-type material, and we can find the equilibrium hole concentration as

$$p_0 = n_i \exp(\frac{E_i - E_F}{kT}) = 1.45 \times 10^{10} \exp(\frac{0.38}{0.026}) \approx 3.23 \times 10^{16} \text{ 1/cm}^3 \tag{7}$$

We can then calculate the equilibrium electron concentration by the Law of Mass Action:

$$n_0 = \frac{n_i^2}{p_0} = \frac{2.1 \times 10^{20}}{3.23 \times 10^{16}} \approx 6.5 \times 10^3 \text{ 1/cm}^3 \tag{8}$$

Note that we could also arrive at this result by using the E_F-related expression for n_0:

$$n_0 = n_i \exp(\frac{E_F - E_i}{kT}) = 1.45 \times 10^{10} \exp(\frac{\mathbf{-0.38}}{0.026}) = 1.45 \times 10^{10} \exp(-14.62) \approx 6.5 \times 10^3 \ 1/\text{cm}^3 \quad (9)$$

As an exercise, go back to Problem 1 and show that you could use the expression in Eqn. 7 to calculate p_0 in that problem after the calculation of E_F in Eqn. 5.

3. Consider a silicon crystal at room temperature, doped with both donor and acceptor atoms so that $N_D = 2 \times 10^{15}$ 1/cm^3 and $N_A = 1 \times 10^{15}$ 1/cm^3. What type of material would this yield? Find the location of the Fermi level.

Here, $N_D > N_A$, so with more donors than acceptors this will be an n-type crystal.

$$n_0 \approx N_D - N_A = 1 \times 10^{15} \text{ 1/cm}^3 \tag{10}$$

Then

$$p_0 = \frac{n_i^2}{n_0} = \frac{2.1 \times 10^{20}}{1 \times 10^{15}} = 2.1 \times 10^5 \text{ 1/cm}^3 \tag{11}$$

and

$$(E_F - E_i) = kT \ln(\frac{n_0}{n_i}) = 0.026 \ln(\frac{1 \times 10^{15}}{1.45 \times 10^{10}}) = 0.026 \cdot 11.14 = 0.29 \text{ eV} \tag{12}$$

The band energy diagram will be as in Figure 3.

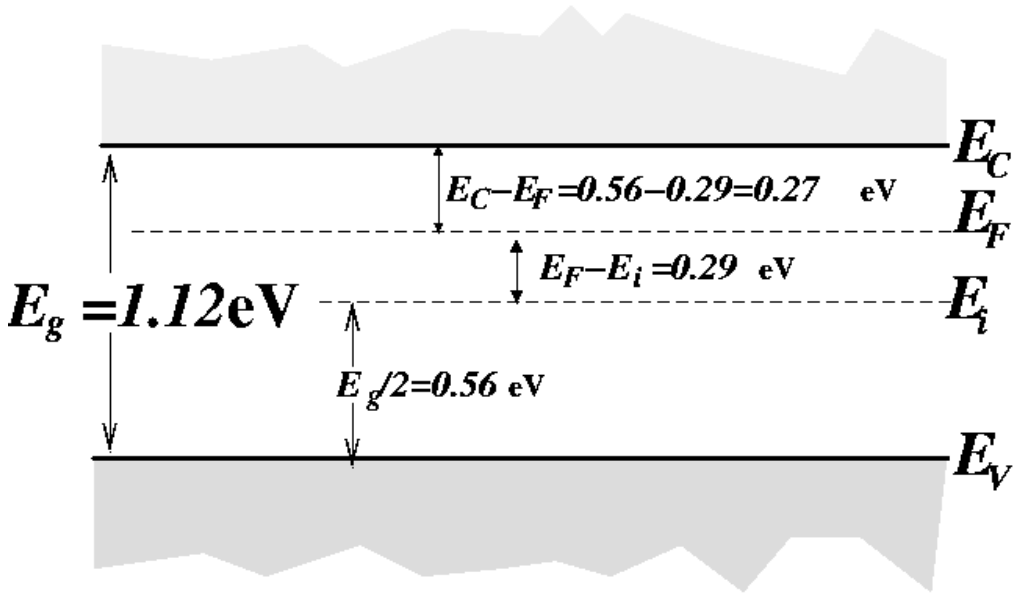


Figure 3: The energy band diagram for the crystal presented in Problem 3.